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Last login: Sun Jun 28 06:36:19 on ttys001
stephenrajva@Stephens-MacBook-Air-4 ~ % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/DragonFly"
echo "==== folder contents ====="
ls -la
echo ""
echo "==== PPN=1 dragonfly config ====="
cat dfly_ppn1.py 2>/dev/null || { echo "dfly_ppn1.py not here - listing:"; ls *.py 2>/dev/null; }
echo ""
echo "==== sweep runner ====="
cat run_dfly_sweep.sh 2>/dev/null || cat run_minimd_dfly_sweep.sh 2>/dev/null || { echo "runner not found - listing .sh files:"; ls *.sh 2>/dev/null; }
==== folder contents =====
total 20080
drwxr-xr-x  29 stephenrajva  staff      928 25 Jun 10:12 .
drwxr-xr-x   9 stephenrajva  staff     288 24 Jun 23:05 ..
-rw-r--r--@  1 stephenrajva  staff  10244 25 Jun 00:49 .DS_Store
-rw-r--r--@  1 stephenrajva  staff   6834 24 Jun 19:42 analyze_universal.py
-rw-r--r--@  1 stephenrajva  staff   4575 24 Jun 19:39 dfly_ppn1.py
-rw-r--r--@  1 stephenrajva  staff  51875 17 Jun 15:05 DragonFly Comparison-Treminal .pdf
-rw-r--r--  1 stephenrajva  staff    113 17 Jun 14:54 dragonfly_comparison.csv
-rw-r--r--  1 stephenrajva  staff    220 17 Jun 15:03 dragonfly_summary.csv
-rw-r--r--  1 stephenrajva  staff   2202 17 Jun 15:03 extract_metrics.py
-rw-r--r--@  1 stephenrajva  staff  25180 17 Jun 14:55 latency.png
-rw-r--r--  1 stephenrajva  staff  51677 24 Jun 19:43 MiniFE_Dragonfly_bits.png
-rw-r--r--  1 stephenrajva  staff  48949 24 Jun 19:43 MiniFE_Dragonfly_idle_pct.png
-rw-r--r--  1 stephenrajva  staff  55220 24 Jun 19:43 MiniFE_Dragonfly_latency_ns.png
-rw-r--r--  1 stephenrajva  staff  51976 24 Jun 19:43 MiniFE_Dragonfly_packets.png
-rw-r--r--  1 stephenrajva  staff  29649 24 Jun 19:43 MiniFE_Dragonfly_stalls.png
-rw-r--r--  1 stephenrajva  staff  53728 24 Jun 19:43 MiniFE_Dragonfly_throughput copy.png
-rw-r--r--  1 stephenrajva  staff  53728 24 Jun 19:43 MiniFE_Dragonfly_throughput.png
-rw-r--r--@  1 stephenrajva  staff  24060 17 Jun 14:55 packets.png
-rw-r--r--  1 stephenrajva  staff    720 17 Jun 14:55 plot_dragonfly.py
drwxr-xr-x   8 stephenrajva  staff    256 24 Jun 19:40 results
-rwxr-xr-x@  1 stephenrajva  staff    455 24 Jun 19:39 run_dfly_sweep.sh
-rw-r--r--@  1 stephenrajva  staff 1737492 24 Jun 20:06 Screenshot 2026-06-24 at 8.06.52 PM.png
-rw-r--r--@  1 stephenrajva  staff 1775280 24 Jun 20:07 Screenshot 2026-06-24 at 8.07.04 PM.png
-rw-r--r--@  1 stephenrajva  staff 1794733 24 Jun 20:07 Screenshot 2026-06-24 at 8.07.15 PM.png
-rw-r--r--@  1 stephenrajva  staff 1697098 24 Jun 20:07 Screenshot 2026-06-24 at 8.07.34 PM.png
-rw-r--r--@  1 stephenrajva  staff 1921296 24 Jun 20:07 Screenshot 2026-06-24 at 8.07.52 PM.png
-rw-r--r--@  1 stephenrajva  staff  764786 24 Jun 20:08 Screenshot 2026-06-24 at 8.08.36 PM.png
-rw-r--r--@  1 stephenrajva  staff   25220 17 Jun 14:55 sendbits.png
-rw-r--r--@  1 stephenrajva  staff   24635 16 Jun 20:56 Untitled.pdf

==== PPN=1 dragonfly config =====
#!/usr/bin/env python
# =====
# dfly_ppn1.py - MiniFE on a Dragonfly, PPN (ranks per node) = 1
# =====
# Dragonfly node count = hosts_per_router * routers_per_group * num_groups.
# To keep PPN = 1 we FIX hosts_per_router = 1, so each router holds one node,
# and we grow the GROUP structure to reach the target size:
#
#      size  routers_per_group  num_groups  nodes
#      4      2                2          4
#      16     4                4         16
#      64     8                8         64
#
# job["size"] = nodes -> PPN = nodes/nodes = 1.
#
# USAGE: export DFLY_SIZE="16" (then) sst dfly_ppn1.py
# The runner loops over all sizes.
# =====

import sst, os
from sst.merlin.base import *
from sst.merlin.endpoint import *
from sst.merlin.interface import *
from sst.merlin.topology import *
from sst.ember import *

# size label -> (routers_per_group, num_groups), all with hosts_per_router = 1
SIZE_TABLE = {

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    "4": (2, 2),
    "16": (4, 4),
    "64": (8, 8),
}

size_label = os.environ.get("DFLY_SIZE", "16")
out_dir = os.environ.get("STATS_DIR", ".")
if size_label not in SIZE_TABLE:
    raise SystemExit(f"DFLY_SIZE must be one of {list(SIZE_TABLE)}, got '{size_label}'")

hosts_per_router = 1 # FIXED -> PPN = 1
routers_per_group, num_groups = SIZE_TABLE[size_label]
num_nodes = hosts_per_router * routers_per_group * num_groups
num_ranks = num_nodes # PPN = 1

PlatformDefinition.setCurrentPlatform("firefly-defaults")

topo = topoDragonFly()
topo.hosts_per_router = hosts_per_router
topo.routers_per_group = routers_per_group
topo.num_groups = num_groups
topo.intergroup_links = 2 # kept from your original configs
topo.link_latency = "25ns"
topo.algorithm = "minimal"

router = hr_router()
router.link_bw = "12GB/s"
router.flit_size = "16B"
router.xbar_bw = "20GB/s"
router.input_latency = "30ns"
router.output_latency = "30ns"
router.input_buf_size = "32kB"
router.output_buf_size = "32kB"
router.num_vns = 2
router.xbar_arb = "merlin.xbar_arb_lru"
topo.router = router

nic = ReorderLinkControl()
nic.link_bw = "12GB/s"
nic.input_buf_size = "16kB"
nic.output_buf_size = "16kB"

system = System()
system.setTopology(topo)

# MiniFE motifs (same collectives as mesh/torus so the comparison is fair)
motifs = [
    {"pattern": "Allreduce", "params": "iterations=92 count=1 compute=69ns"},
    {"pattern": "Bcast", "params": "root=0 iterations=2 count=7 compute=1323ns"},
    {"pattern": "Reduce", "params": "root=0 iterations=2 count=2 compute=450ns"},
    {"pattern": "Allgather", "params": "iterations=3 count=3 compute=243ns verify=0"},
    {"pattern": "Barrier", "params": "iterations=99 compute=75ns"},
]

candidates = [1, num_nodes // 2, num_nodes - 1]
partners = sorted({p for p in candidates if 0 < p < num_ranks})
for b in partners:
    motifs.append({"pattern": "PingPong",
                  "params": f"rank2={b} iterations=35 messageSize=256 compute=3ns"})

ep = EmberMPIJob(0, num_ranks)
ep.network_interface = nic
ep.addMotif("Init")
for m in motifs:
    ep.addMotif(f"{m['pattern']} {m['params']}")
ep.addMotif("Fini")
ep.nic.nic2host_lat = "100ns"

system.allocateNodes(ep, "linear")
system.build()

csv_path = os.path.join(out_dir, f"dragonfly_{size_label}_ppn1_nodes{num_nodes}_stats.csv")
sst.setStatisticLoadLevel(10)
sst.setStatisticOutput("sst.statOutputCSV", {"filepath": csv_path, "separator": ", "})

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sst.enableAllStatisticsForComponentType("merlin.hr_router")
sst.enableAllStatisticsForComponentType("merlin.linkcontrol")
sst.enableAllStatisticsForComponentType("ember.EmberEngine", {"MotifLog": "1"})
sst.enableAllStatisticsForComponentType("ember.nic")

print("=====")
print(f" Topology          : Dragonfly (size {size_label})")
print(f" hosts_per_router   : {hosts_per_router}")
print(f" routers_per_group   : {routers_per_group}")
print(f" num_groups          : {num_groups}")
print(f" Total nodes         : {num_nodes}")
print(f" Total MPI ranks     : {num_ranks}")
print(f" PPN (ranks/node)    : {num_ranks // num_nodes}    <-- should be 1")
print(f" Stats CSV          : {csv_path}")
print("=====")

===== sweep runner =====
#!/usr/bin/env bash
set -e
SCRIPT="$(pwd)/dfly_ppn1.py"
OUT_DIR="$(pwd)/results"
# Dragonfly sizes (nodes), all with hosts_per_router=1 -> PPN=1
SIZES=("4" "16" "64")
mkdir -p "$OUT_DIR"
for S in "${SIZES[@]}; do
    echo "=== Running dragonfly size $S (PPN=1) ==="
    export DFLY_SIZE="$S"
    export STATS_DIR="$OUT_DIR"
    sst "$SCRIPT" 2>&1 | tee "$OUT_DIR/dragonfly_${S}_ppn1_run.log"
done
echo "All dragonfly runs finished. Results in: $OUT_DIR"
stephenrajva@Stephens-MacBook-Air-4 DragonFly % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_S
timulator/MiniFE/DragonFly"
mkdir -p PPN_2
ls -l ~/Downloads/dfly_ppn2.py ~/Downloads/run_dfly_sweep_ppn2.sh 2>/dev/null
ls -l PPN_2/
-rw-r--r--@ 1 stephenrajva  staff   4879 28 Jun 06:47 /Users/stephenrajva/Downloads/dfly_ppn2.py
-rw-r--r--@ 1 stephenrajva  staff    472 28 Jun 06:47 /Users/stephenrajva/Downloads/run_dfly_sweep_pp
n2.sh
total 0
stephenrajva@Stephens-MacBook-Air-4 DragonFly % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_S
timulator/MiniFE/DragonFly"
mv ~/Downloads/dfly_ppn2.py ~/Downloads/run_dfly_sweep_ppn2.sh PPN_2/
ls -l PPN_2/
total 24
-rw-r--r--@ 1 stephenrajva  staff   4879 28 Jun 06:47 dfly_ppn2.py
-rw-r--r--@ 1 stephenrajva  staff    472 28 Jun 06:47 run_dfly_sweep_ppn2.sh
stephenrajva@Stephens-MacBook-Air-4 DragonFly % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_S
timulator/MiniFE/DragonFly/PPN_2"
chmod +x run_dfly_sweep_ppn2.sh
export DFLY_SIZE="4"
export STATS_DIR="./results"
sst dfly_ppn2.py
=====
Topology          : Dragonfly (size 4)
hosts_per_router   : 2
routers_per_group   : 2
num_groups          : 2
Total routers      : 4
Total nodes        : 8
Total MPI ranks     : 8
PPN (ranks/node)    : 2    <-- should be 2
Stats CSV          : ./results/dragonfly_4_ppn2_nodes8_stats.csv
=====
Allreduce: ranks 8, loop 92, 1 double(s), latency 5.507 us
generate
510115133:0:0:EmberEngine:Reduce:generate:37: rank=0 size=8
generate
511458333:0:1:EmberEngine:Reduce:generate:37: rank=1 size=8
generate
511674333:0:2:EmberEngine:Reduce:generate:37: rank=2 size=8
generate
512257533:0:3:EmberEngine:Reduce:generate:37: rank=3 size=8

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generate
512429933:0:5:EmberEngine:Reduce:generate:37: rank=5 size=8
generate
512430733:0:4:EmberEngine:Reduce:generate:37: rank=4 size=8
generate
512646733:0:6:EmberEngine:Reduce:generate:37: rank=6 size=8
generate
Bcast: ranks 8, loop 2, bytes 56, latency 1.893 us
513057133:0:7:EmberEngine:Reduce:generate:37: rank=7 size=8
Allgather: ranks 8, loop 3, bytes 12, latency 4.771 us
Barrier: ranks 8, loop 99, latency 5.512 us
1099297933:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=1 messageSize=256 iterations=35
1100141133:0:1:EmberEngine:PingPong:generate:65: rank=1 size=8
1101113133:0:4:EmberEngine:PingPong:generate:65: rank=4 size=8
1101740333:0:7:EmberEngine:PingPong:generate:65: rank=7 size=8
PingPong: otherRank 1, total time 63.474 us, loop 35, bufLen 256, latency 0.907 us. bandwidth 0.2823
20 GB/s
1162771533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=4 messageSize=256 iterations=35
PingPong: otherRank 4, total time 69.272 us, loop 35, bufLen 256, latency 0.990 us. bandwidth 0.2586
90 GB/s
1232043533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=7 messageSize=256 iterations=35
PingPong: otherRank 7, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.2379
18 GB/s
Simulation is complete, simulated time: 1.31194 ms
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimu
lator/MiniFE/DragonFly/PPN_2"
./run_dfly_sweep_ppn2.sh
=== Running dragonfly size 4 (PPN=2) ===
=====
Topology          : Dragonfly (size 4)
hosts_per_router  : 2
routers_per_group : 2
num_groups        : 2
Total routers     : 4
Total nodes       : 8
Total MPI ranks   : 8
PPN (ranks/node)  : 2  <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/DragonFly/PPN_
2/results/dragonfly_4_ppn2_nodes8_stats.csv
=====
Allreduce: ranks 8, loop 92, 1 double(s), latency 5.507 us
generate
510115133:0:0:EmberEngine:Reduce:generate:37: rank=0 size=8
generate
511458333:0:1:EmberEngine:Reduce:generate:37: rank=1 size=8
generate
511674333:0:2:EmberEngine:Reduce:generate:37: rank=2 size=8
generate
512257533:0:3:EmberEngine:Reduce:generate:37: rank=3 size=8
generate
512429933:0:5:EmberEngine:Reduce:generate:37: rank=5 size=8
generate
512430733:0:4:EmberEngine:Reduce:generate:37: rank=4 size=8
generate
512646733:0:6:EmberEngine:Reduce:generate:37: rank=6 size=8
generate
Bcast: ranks 8, loop 2, bytes 56, latency 1.893 us
513057133:0:7:EmberEngine:Reduce:generate:37: rank=7 size=8
Allgather: ranks 8, loop 3, bytes 12, latency 4.771 us
Barrier: ranks 8, loop 99, latency 5.512 us
1099297933:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=1 messageSize=256 iterations=35
1100141133:0:1:EmberEngine:PingPong:generate:65: rank=1 size=8
1101113133:0:4:EmberEngine:PingPong:generate:65: rank=4 size=8
1101740333:0:7:EmberEngine:PingPong:generate:65: rank=7 size=8
PingPong: otherRank 1, total time 63.474 us, loop 35, bufLen 256, latency 0.907 us. bandwidth 0.2823
20 GB/s
1162771533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=4 messageSize=256 iterations=35
PingPong: otherRank 4, total time 69.272 us, loop 35, bufLen 256, latency 0.990 us. bandwidth 0.2586

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90 GB/s
1232043533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=8
rank2=7 messageSize=256 iterations=35
PingPong: otherRank 7, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.2379
18 GB/s
Simulation is complete, simulated time: 1.31194 ms
=== Running dragonfly size 16 (PPN=2) ===
=====
Topology      : Dragonfly (size 16)
hosts_per_router : 2
routers_per_group : 4
num_groups      : 4
Total routers    : 16
Total nodes      : 32
Total MPI ranks   : 32
PPN (ranks/node) : 2  <-- should be 2
Stats CSV        : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/DragonFly/PPN_
2/results/dragonfly_16_ppn2_nodes32_stats.csv
=====
Allreduce: ranks 32, loop 92, 1 double(s), latency 9.724 us
generate
898075133:0:0:EmberEngine:Reduce:generate:37: rank=0 size=32
generate
899418333:0:1:EmberEngine:Reduce:generate:37: rank=1 size=32
generate
899634333:0:2:EmberEngine:Reduce:generate:37: rank=2 size=32
generate
900347933:0:3:EmberEngine:Reduce:generate:37: rank=3 size=32
generate
900478133:0:4:EmberEngine:Reduce:generate:37: rank=4 size=32
generate
900563933:0:5:EmberEngine:Reduce:generate:37: rank=5 size=32
generate
900694133:0:6:EmberEngine:Reduce:generate:37: rank=6 size=32
generate
901277533:0:7:EmberEngine:Reduce:generate:37: rank=7 size=32
generate
901494133:0:8:EmberEngine:Reduce:generate:37: rank=8 size=32
generate
901494133:0:9:EmberEngine:Reduce:generate:37: rank=9 size=32
generate
901579933:0:11:EmberEngine:Reduce:generate:37: rank=11 size=32
generate
901710733:0:10:EmberEngine:Reduce:generate:37: rank=10 size=32
generate
901796533:0:12:EmberEngine:Reduce:generate:37: rank=12 size=32
generate
901796533:0:13:EmberEngine:Reduce:generate:37: rank=13 size=32
generate
901926733:0:14:EmberEngine:Reduce:generate:37: rank=14 size=32
generate
902163533:0:17:EmberEngine:Reduce:generate:37: rank=17 size=32
generate
902249533:0:15:EmberEngine:Reduce:generate:37: rank=15 size=32
generate
902249933:0:16:EmberEngine:Reduce:generate:37: rank=16 size=32
generate
902336333:0:19:EmberEngine:Reduce:generate:37: rank=19 size=32
generate
902380333:0:18:EmberEngine:Reduce:generate:37: rank=18 size=32
generate
902380333:0:20:EmberEngine:Reduce:generate:37: rank=20 size=32
generate
902421933:0:23:EmberEngine:Reduce:generate:37: rank=23 size=32
generate
902552333:0:24:EmberEngine:Reduce:generate:37: rank=24 size=32
generate
902552333:0:25:EmberEngine:Reduce:generate:37: rank=25 size=32
generate
902553133:0:21:EmberEngine:Reduce:generate:37: rank=21 size=32
generate
902638733:0:27:EmberEngine:Reduce:generate:37: rank=27 size=32
generate

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902682733:0:22:EmberEngine:Reduce:generate:37: rank=22 size=32
generate
902682733:0:28:EmberEngine:Reduce:generate:37: rank=28 size=32
generate
902769133:0:29:EmberEngine:Reduce:generate:37: rank=29 size=32
generate
902778733:0:26:EmberEngine:Reduce:generate:37: rank=26 size=32
generate
902898733:0:30:EmberEngine:Reduce:generate:37: rank=30 size=32
generate
Bcast: ranks 32, loop 2, bytes 56, latency 1.893 us
903135533:0:31:EmberEngine:Reduce:generate:37: rank=31 size=32
Allgather: ranks 32, loop 3, bytes 12, latency 6.382 us
Barrier: ranks 32, loop 99, latency 9.753 us
1916967533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=1 messageSize=256 iterations=35
1917810733:0:1:EmberEngine:PingPong:generate:65: rank=1 size=32
1920641933:0:16:EmberEngine:PingPong:generate:65: rank=16 size=32
1921528333:0:31:EmberEngine:PingPong:generate:65: rank=31 size=32
PingPong: otherRank 1, total time 63.474 us, loop 35, bufLen 256, latency 0.907 us. bandwidth 0.2823
20 GB/s
1980441133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=16 messageSize=256 iterations=35
PingPong: otherRank 16, total time 69.272 us, loop 35, bufLen 256, latency 0.990 us. bandwidth 0.258
690 GB/s
2049713133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=32
rank2=31 messageSize=256 iterations=35
PingPong: otherRank 31, total time 81.368 us, loop 35, bufLen 256, latency 1.162 us. bandwidth 0.220
234 GB/s
Simulation is complete, simulated time: 2.13981 ms
=== Running dragonfly size 64 (PPN=2) ===
=====
Topology          : Dragonfly (size 64)
hosts_per_router  : 2
routers_per_group : 8
num_groups        : 8
Total routers     : 64
Total nodes       : 128
Total MPI ranks   : 128
PPN (ranks/node)  : 2  <-- should be 2
Stats CSV         : /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/DragonFly/PPN_
2/results/dragonfly_64_ppn2_nodes128_stats.csv
=====
Allreduce: ranks 128, loop 92, 1 double(s), latency 13.953 us
generate
1287165533:0:0:EmberEngine:Reduce:generate:37: rank=0 size=128
generate
1288508733:0:1:EmberEngine:Reduce:generate:37: rank=1 size=128
generate
1288724733:0:2:EmberEngine:Reduce:generate:37: rank=2 size=128
generate
1289438333:0:3:EmberEngine:Reduce:generate:37: rank=3 size=128
generate
1289568533:0:4:EmberEngine:Reduce:generate:37: rank=4 size=128
generate
1289654333:0:5:EmberEngine:Reduce:generate:37: rank=5 size=128
generate
1289784533:0:6:EmberEngine:Reduce:generate:37: rank=6 size=128
generate
1290367933:0:7:EmberEngine:Reduce:generate:37: rank=7 size=128
generate
1290498133:0:8:EmberEngine:Reduce:generate:37: rank=8 size=128
generate
1290498133:0:9:EmberEngine:Reduce:generate:37: rank=9 size=128
generate
1290583933:0:11:EmberEngine:Reduce:generate:37: rank=11 size=128
generate
1290628333:0:10:EmberEngine:Reduce:generate:37: rank=10 size=128
generate
1290714133:0:12:EmberEngine:Reduce:generate:37: rank=12 size=128
generate
1290714133:0:13:EmberEngine:Reduce:generate:37: rank=13 size=128
generate

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1290844333:0:14:EmberEngine:Reduce:generate:37: rank=14 size=128
generate
1291297533:0:15:EmberEngine:Reduce:generate:37: rank=15 size=128
generate
1291514133:0:16:EmberEngine:Reduce:generate:37: rank=16 size=128
generate
1291514133:0:17:EmberEngine:Reduce:generate:37: rank=17 size=128
generate
1291599933:0:23:EmberEngine:Reduce:generate:37: rank=23 size=128
generate
1291600533:0:19:EmberEngine:Reduce:generate:37: rank=19 size=128
generate
1291730733:0:20:EmberEngine:Reduce:generate:37: rank=20 size=128
generate
1291730733:0:21:EmberEngine:Reduce:generate:37: rank=21 size=128
generate
1291740333:0:18:EmberEngine:Reduce:generate:37: rank=18 size=128
generate
1291816533:0:25:EmberEngine:Reduce:generate:37: rank=25 size=128
generate
1291816533:0:27:EmberEngine:Reduce:generate:37: rank=27 size=128
generate
1291826133:0:24:EmberEngine:Reduce:generate:37: rank=24 size=128
generate
1291860333:0:22:EmberEngine:Reduce:generate:37: rank=22 size=128
generate
1291946733:0:28:EmberEngine:Reduce:generate:37: rank=28 size=128
generate
1291946733:0:29:EmberEngine:Reduce:generate:37: rank=29 size=128
generate
1291956333:0:26:EmberEngine:Reduce:generate:37: rank=26 size=128
generate
1292076333:0:30:EmberEngine:Reduce:generate:37: rank=30 size=128
generate
1292399933:0:31:EmberEngine:Reduce:generate:37: rank=31 size=128
generate
1292443733:0:33:EmberEngine:Reduce:generate:37: rank=33 size=128
generate
1292530133:0:32:EmberEngine:Reduce:generate:37: rank=32 size=128
generate
1292616533:0:35:EmberEngine:Reduce:generate:37: rank=35 size=128
generate
1292660333:0:34:EmberEngine:Reduce:generate:37: rank=34 size=128
generate
1292707933:0:47:EmberEngine:Reduce:generate:37: rank=47 size=128
generate
1292711733:0:39:EmberEngine:Reduce:generate:37: rank=39 size=128
generate
1292746733:0:36:EmberEngine:Reduce:generate:37: rank=36 size=128
generate
1292756333:0:40:EmberEngine:Reduce:generate:37: rank=40 size=128
generate
1292832533:0:51:EmberEngine:Reduce:generate:37: rank=51 size=128
generate
1292833133:0:41:EmberEngine:Reduce:generate:37: rank=41 size=128
generate
1292833133:0:43:EmberEngine:Reduce:generate:37: rank=43 size=128
generate
1292842133:0:48:EmberEngine:Reduce:generate:37: rank=48 size=128
generate
1292842133:0:49:EmberEngine:Reduce:generate:37: rank=49 size=128
generate
1292842733:0:37:EmberEngine:Reduce:generate:37: rank=37 size=128
generate
1292918933:0:55:EmberEngine:Reduce:generate:37: rank=55 size=128
generate
1292962733:0:44:EmberEngine:Reduce:generate:37: rank=44 size=128
generate
1292962733:0:45:EmberEngine:Reduce:generate:37: rank=45 size=128
generate
1292962733:0:52:EmberEngine:Reduce:generate:37: rank=52 size=128
generate
1292972333:0:42:EmberEngine:Reduce:generate:37: rank=42 size=128

generate
1292972333:0:56:EmberEngine:Reduce:generate:37: rank=56 size=128
generate
1292981933:0:38:EmberEngine:Reduce:generate:37: rank=38 size=128
generate
1293049133:0:57:EmberEngine:Reduce:generate:37: rank=57 size=128
generate
1293049133:0:59:EmberEngine:Reduce:generate:37: rank=59 size=128
generate
1293058733:0:53:EmberEngine:Reduce:generate:37: rank=53 size=128
generate
1293068333:0:50:EmberEngine:Reduce:generate:37: rank=50 size=128
generate
1293092333:0:46:EmberEngine:Reduce:generate:37: rank=46 size=128
generate
1293178733:0:60:EmberEngine:Reduce:generate:37: rank=60 size=128
generate
1293178733:0:61:EmberEngine:Reduce:generate:37: rank=61 size=128
generate
1293188333:0:58:EmberEngine:Reduce:generate:37: rank=58 size=128
generate
1293197933:0:54:EmberEngine:Reduce:generate:37: rank=54 size=128
generate
1293285933:0:67:EmberEngine:Reduce:generate:37: rank=67 size=128
generate
1293308333:0:62:EmberEngine:Reduce:generate:37: rank=62 size=128
generate
1293371933:0:63:EmberEngine:Reduce:generate:37: rank=63 size=128
generate
1293372333:0:64:EmberEngine:Reduce:generate:37: rank=64 size=128
generate
1293372333:0:65:EmberEngine:Reduce:generate:37: rank=65 size=128
generate
1293416333:0:66:EmberEngine:Reduce:generate:37: rank=66 size=128
generate
1293416333:0:68:EmberEngine:Reduce:generate:37: rank=68 size=128
generate
1293416333:0:69:EmberEngine:Reduce:generate:37: rank=69 size=128
generate
1293458733:0:71:EmberEngine:Reduce:generate:37: rank=71 size=128
generate
1293502733:0:72:EmberEngine:Reduce:generate:37: rank=72 size=128
generate
1293545933:0:70:EmberEngine:Reduce:generate:37: rank=70 size=128
generate
1293549933:0:95:EmberEngine:Reduce:generate:37: rank=95 size=128
generate
1293553933:0:79:EmberEngine:Reduce:generate:37: rank=79 size=128
generate
1293589133:0:73:EmberEngine:Reduce:generate:37: rank=73 size=128
generate
1293589133:0:87:EmberEngine:Reduce:generate:37: rank=87 size=128
generate
1293598733:0:81:EmberEngine:Reduce:generate:37: rank=81 size=128
generate
1293632333:0:88:EmberEngine:Reduce:generate:37: rank=88 size=128
generate
1293641933:0:82:EmberEngine:Reduce:generate:37: rank=82 size=128
generate
1293674733:0:103:EmberEngine:Reduce:generate:37: rank=103 size=128
generate
1293675533:0:83:EmberEngine:Reduce:generate:37: rank=83 size=128
generate
1293680333:0:96:EmberEngine:Reduce:generate:37: rank=96 size=128
generate
1293684333:0:97:EmberEngine:Reduce:generate:37: rank=97 size=128
generate
1293684333:0:99:EmberEngine:Reduce:generate:37: rank=99 size=128
generate
1293685133:0:75:EmberEngine:Reduce:generate:37: rank=75 size=128
generate
1293685133:0:80:EmberEngine:Reduce:generate:37: rank=80 size=128
generate

1293718733:0:74:EmberEngine:Reduce:generate:37: rank=74 size=128
generate
1293718733:0:105:EmberEngine:Reduce:generate:37: rank=105 size=128
generate
1293728333:0:98:EmberEngine:Reduce:generate:37: rank=98 size=128
generate
1293761133:0:111:EmberEngine:Reduce:generate:37: rank=111 size=128
generate
1293805133:0:89:EmberEngine:Reduce:generate:37: rank=89 size=128
generate
1293805133:0:91:EmberEngine:Reduce:generate:37: rank=91 size=128
generate
1293805133:0:104:EmberEngine:Reduce:generate:37: rank=104 size=128
generate
1293814733:0:76:EmberEngine:Reduce:generate:37: rank=76 size=128
generate
1293814733:0:84:EmberEngine:Reduce:generate:37: rank=84 size=128
generate
1293814733:0:85:EmberEngine:Reduce:generate:37: rank=85 size=128
generate
1293814733:0:100:EmberEngine:Reduce:generate:37: rank=100 size=128
generate
1293814733:0:107:EmberEngine:Reduce:generate:37: rank=107 size=128
generate
1293814733:0:113:EmberEngine:Reduce:generate:37: rank=113 size=128
generate
1293824333:0:77:EmberEngine:Reduce:generate:37: rank=77 size=128
generate
1293848333:0:106:EmberEngine:Reduce:generate:37: rank=106 size=128
generate
1293857933:0:114:EmberEngine:Reduce:generate:37: rank=114 size=128
generate
1293891533:0:112:EmberEngine:Reduce:generate:37: rank=112 size=128
generate
1293901133:0:115:EmberEngine:Reduce:generate:37: rank=115 size=128
generate
1293910733:0:101:EmberEngine:Reduce:generate:37: rank=101 size=128
generate
1293910733:0:119:EmberEngine:Reduce:generate:37: rank=119 size=128
generate
1293934733:0:92:EmberEngine:Reduce:generate:37: rank=92 size=128
generate
1293934733:0:93:EmberEngine:Reduce:generate:37: rank=93 size=128
generate
1293934733:0:123:EmberEngine:Reduce:generate:37: rank=123 size=128
generate
1293944333:0:90:EmberEngine:Reduce:generate:37: rank=90 size=128
generate
1293953933:0:78:EmberEngine:Reduce:generate:37: rank=78 size=128
generate
1293953933:0:86:EmberEngine:Reduce:generate:37: rank=86 size=128
generate
1294021133:0:121:EmberEngine:Reduce:generate:37: rank=121 size=128
generate
1294030733:0:108:EmberEngine:Reduce:generate:37: rank=108 size=128
generate
1294030733:0:116:EmberEngine:Reduce:generate:37: rank=116 size=128
generate
1294030733:0:117:EmberEngine:Reduce:generate:37: rank=117 size=128
generate
1294040333:0:109:EmberEngine:Reduce:generate:37: rank=109 size=128
generate
1294040333:0:120:EmberEngine:Reduce:generate:37: rank=120 size=128
generate
1294049933:0:102:EmberEngine:Reduce:generate:37: rank=102 size=128
generate
1294064333:0:94:EmberEngine:Reduce:generate:37: rank=94 size=128
generate
1294150733:0:124:EmberEngine:Reduce:generate:37: rank=124 size=128
generate
1294150733:0:125:EmberEngine:Reduce:generate:37: rank=125 size=128
generate
1294160333:0:122:EmberEngine:Reduce:generate:37: rank=122 size=128

```

generate
1294169933:0:110:EmberEngine:Reduce:generate:37: rank=110 size=128
generate
1294169933:0:118:EmberEngine:Reduce:generate:37: rank=118 size=128
generate
Bcast: ranks 128, loop 2, bytes 56, latency 1.893 us
1294257933:0:127:EmberEngine:Reduce:generate:37: rank=127 size=128
generate
1294280333:0:126:EmberEngine:Reduce:generate:37: rank=126 size=128
Allgather: ranks 128, loop 3, bytes 12, latency 8.679 us
Barrier: ranks 128, loop 99, latency 13.946 us
2733085933:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=1 messageSize=256 iterations=35
2733929133:0:1:EmberEngine:PingPong:generate:65: rank=1 size=128
2738792333:0:64:EmberEngine:PingPong:generate:65: rank=64 size=128
2739678733:0:127:EmberEngine:PingPong:generate:65: rank=127 size=128
PingPong: otherRank 1, total time 63.474 us, loop 35, bufLen 256, latency 0.907 us. bandwidth 0.2823
20 GB/s
2796559533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=64 messageSize=256 iterations=35
PingPong: otherRank 64, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237
918 GB/s
2871879533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=128
rank2=127 messageSize=256 iterations=35
PingPong: otherRank 127, total time 81.368 us, loop 35, bufLen 256, latency 1.162 us. bandwidth 0.22
0234 GB/s
Simulation is complete, simulated time: 2.96606 ms
All PPN=2 dragonfly runs finished. Results in: /Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimu
lator/MiniFE/DragonFly/PPN_2/results
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % ls -l results/*.csv
-rw-r--r--  1 stephenrajva  staff   67357 28 Jun 06:48 results/dragonfly_16_ppn2_nodes32_stats.csv
-rw-r--r--  1 stephenrajva  staff  11624 28 Jun 06:48 results/dragonfly_4_ppn2_nodes8_stats.csv
-rw-r--r--  1 stephenrajva  staff  407691 28 Jun 06:48 results/dragonfly_64_ppn2_nodes128_stats.csv
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimu
lator/MiniFE/DragonFly/PPN_2"
cp ../analyze_universal.py .
python3 - << 'EOF'
import re
f = "analyze_universal.py"
src = open(f).read()
src = re.sub(r'^STUDY\s*=\s*$', 'STUDY = "MiniFE_Dragonfly_PPN2"', src, count=1, flags=re.M)
src = re.sub(r'^TOPOLOGY\s*=\s*$', 'TOPOLOGY = "other"', src, count=1, flags=re.M)
runs = '''RUNS = [
    ("4", (2, 2), "dragonfly_4_ppn2_nodes8_stats.csv"),
    ("16", (4, 4), "dragonfly_16_ppn2_nodes32_stats.csv"),
    ("64", (8, 8), "dragonfly_64_ppn2_nodes128_stats.csv"),
]'''
src = re.sub(r'^RUNS\s*=\s*\s*$', runs, src, count=1, flags=re.M | re.S)
src = src.replace('PPN = 1', 'PPN = 2').replace('PPN=1', 'PPN=2')
open(f, "w").write(src)
print("Config updated:")
for line in src.splitlines():
    if line.startswith(("STUDY", "TOPOLOGY", "RUNS")) or 'ppn2_nodes' in line:
        print(line)
EOF
Config updated:
STUDY = "MiniFE_Dragonfly_PPN2"
TOPOLOGY = "other"
RUNS = [
    ("4", (2, 2), "dragonfly_4_ppn2_nodes8_stats.csv"),
    ("16", (4, 4), "dragonfly_16_ppn2_nodes32_stats.csv"),
    ("64", (8, 8), "dragonfly_64_ppn2_nodes128_stats.csv"),
]
stephenrajva@Stephens-MacBook-Air-4 PPN_2 % python3 analyze_universal.py

MiniFE_Dragonfly_PPN2
Size      Latency(ns)  Hops    Thru(GB/s)  Packets  Bits          Idle%  Stalls
-----
4         232.10      -        0.3551     7232     3727296     91.0   0
16        296.53      -        1.0333     40208    17688992    96.5   0
64        327.74      -        3.7924    183626   89989040    98.4   0
Saved: MiniFE_Dragonfly_PPN2_latency_ns.png
(skipping Average hop count: not available for topology 'other')
Saved: MiniFE_Dragonfly_PPN2_throughput.png

```